

CS 171: Discussion Section 3 (Sept 15)

1. Pseudorandom Generators

Let $F, G : \{0, 1\}^n \rightarrow \{0, 1\}^{3n}$ be pseudorandom generators. For each of the functions below, prove or disprove that H is necessarily a pseudorandom generator.

- (a) $H(s_0 s_1 \dots s_{n-1}) := G(s_{n-1} s_{n-2} \dots s_0)$.
- (b) $H(s) := G(s)_{0, \dots, 2n-1}$ (i.e., the first $2n$ bits of $G(s)$).
- (c) $H(s) = G(s) \| F(s)$.

Solution (a) H is a PRG. Suppose, toward a contradiction, that H is not a PRG. Then there is a PPT adversary $\mathcal{A}$ such that

$$|\Pr[\mathcal{A}(U_{3n}) = 1] - \Pr[\mathcal{A}(H(U_n)) = 1]|$$

is non-negligible. Construct a PPT adversary $\mathcal{B}$ against G as follows: on input $t \in \{0, 1\}^{3n}$, output $\mathcal{A}(t)$.

Reversing the bits of a uniform string produces another uniform string. Thus, if $S \xleftarrow{\$} \{0, 1\}^n$, then

$$H(S) = G(S_{n-1} S_{n-2} \dots S_0)$$

has exactly the same distribution as $G(U_n)$. Therefore,

$$\begin{aligned} \Pr[\mathcal{B}(U_{3n}) = 1] &= \Pr[\mathcal{A}(U_{3n}) = 1], \\ \Pr[\mathcal{B}(G(U_n)) = 1] &= \Pr[\mathcal{A}(H(U_n)) = 1]. \end{aligned}$$

Hence $\mathcal{B}$ distinguishes $G(U_n)$ from U_{3n} with the same non-negligible advantage as $\mathcal{A}$, contradicting the pseudorandomness of G .

(b) H is a PRG. Suppose, toward a contradiction, that H is not a PRG. Then there is a PPT adversary $\mathcal{A}$ such that

$$|\Pr[\mathcal{A}(U_{2n}) = 1] - \Pr[\mathcal{A}(H(U_n)) = 1]|$$

is non-negligible. Construct a PPT adversary $\mathcal{B}$ against G as follows: on input $t \in \{0, 1\}^{3n}$, output $\mathcal{A}(t_{0,1,\dots,2n-1})$. The first $2n$ bits of U_{3n} are distributed as U_{2n} , while the first $2n$ bits of $G(U_n)$ are distributed as $H(U_n)$. Therefore,

$$\begin{aligned} \Pr[\mathcal{B}(U_{3n}) = 1] &= \Pr[\mathcal{A}(U_{2n}) = 1], \\ \Pr[\mathcal{B}(G(U_n)) = 1] &= \Pr[\mathcal{A}(H(U_n)) = 1]. \end{aligned}$$

Thus $\mathcal{B}$ breaks the pseudorandomness of G , a contradiction.

(c) H is not necessarily a PRG. As a counterexample, let $F = G$. Then

$$H(s) = G(s) \| G(s),$$

so its first and second halves are always equal. A distinguisher can test whether the two halves are equal. It outputs 1 with probability 1 on $H(U_n)$, but with probability only 2^{-3n} on U_{6n} . Its advantage is therefore $1 - 2^{-3n}$, which is non-negligible. ■

2. Unpredictability and Pseudorandomness

Consider an efficiently computable keyed function $F : \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$. We'll define the following game $\text{Pred}_{\mathcal{A},F}(n)$:

1. Sample $k \leftarrow \{0,1\}^n$ uniformly and set $O := F_k(\cdot)$.
2. $\mathcal{A}$ is given 1^n and oracle access to $O(\cdot)$: it may query any $x \in \{0,1\}^n$ and receives $O(x)$. Let Q be the set of inputs queried by $\mathcal{A}$.
3. $\mathcal{A}$ outputs a prediction $(x^*, y^*) \in \{0,1\}^n \times \{0,1\}^n$.
4. Output 1 if $x^* \notin Q$ and $y^* = O(x^*)$, and 0 otherwise.

We say that F is *unpredictable* if for every probabilistic polynomial-time adversary $\mathcal{A}$:

$$\Pr[\text{Pred}_{\mathcal{A},F}(n) = 1] \leq \text{negl}(n).$$

- (a) Why does the definition require that x^* was never queried?
- (b) Let $F : \{0,1\}^n \times \{0,1\}^n \rightarrow \{0,1\}^n$ be a secure PRF. Prove that F is unpredictable.
- (c) Does unpredictability imply pseudorandomness? Either prove the implication or construct a function family F that is unpredictable but not pseudorandom, and prove both properties.

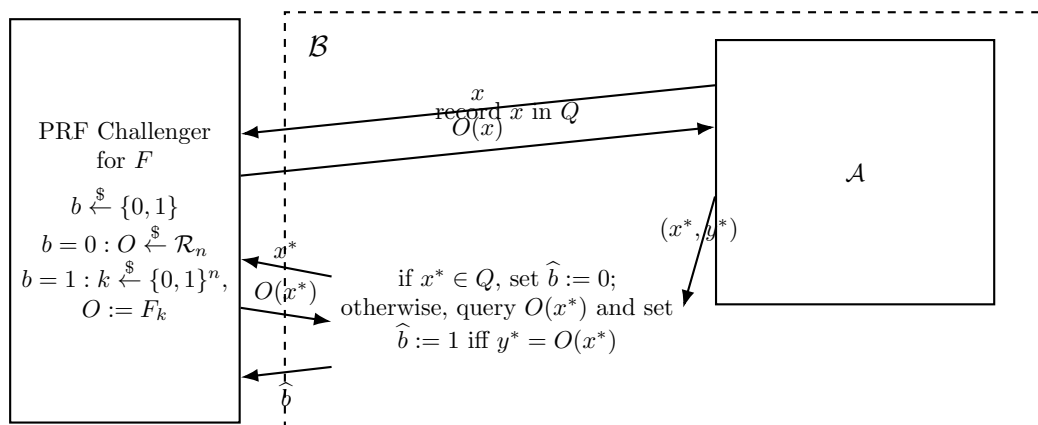
Solution (a) Without this restriction, the definition would be meaningless: no function family could satisfy it. For any F , an adversary could query 0^n , receive $F_k(0^n)$, and then output $(0^n, F_k(0^n))$. It would win with probability 1.

(b) Suppose, toward a contradiction, that F is a secure PRF but is not unpredictable. Then there is a PPT adversary $\mathcal{A}$ whose success probability

$$p(n) := \Pr[\text{Pred}_{\mathcal{A},F}(n) = 1]$$

is non-negligible. We use $\mathcal{A}$ to construct a PPT adversary $\mathcal{B}$ for the PRF game.

Let O be $\mathcal{B}$'s oracle. The PRF challenger chooses a hidden bit b : if $b = 0$, then O is a uniformly random function, and if $b = 1$, then $O = F_k$ for a uniformly random key k . Adversary $\mathcal{B}$ runs $\mathcal{A}$, forwards each of its oracle queries to O , and records the set Q of queried inputs. When $\mathcal{A}$ outputs (x^*, y^*) , $\mathcal{B}$ outputs 0 if $x^* \in Q$. Otherwise, $\mathcal{B}$ queries $O(x^*)$ and outputs 1 if and only if $O(x^*) = y^*$.



There are two cases:

1. If $b = 1$, then $O = F_k$. In this case $\mathcal{A}$ is playing exactly the unpredictability game, so

$$\Pr[\mathcal{B} \text{ outputs } 1 \mid b = 1] = p(n).$$

2. If $b = 0$, then O is a uniformly random function. Conditioned on $x^* \notin Q$, the value $O(x^*)$ is uniform and independent of $\mathcal{A}$'s view. Consequently,

$$\Pr[\mathcal{B} \text{ outputs } 1 \mid b = 0] \leq 2^{-n}.$$

It follows that $\mathcal{B}$'s probability of guessing b correctly is

$$\begin{aligned} \Pr[\text{PRF}_{\mathcal{B}, F}(n) = 1] &= \frac{1}{2} \Pr[\mathcal{B} \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \Pr[\mathcal{B} \text{ outputs } 1 \mid b = 1] \\ &\geq \frac{1}{2}(1 - 2^{-n}) + \frac{1}{2}p(n) \\ &= \frac{1}{2} + \frac{p(n) - 2^{-n}}{2}. \end{aligned}$$

Because $p(n)$ is non-negligible and 2^{-n} is negligible, $\frac{p(n) - 2^{-n}}{2}$ is non-negligible. Thus $\mathcal{B}$ breaks the PRF security of F , a contradiction.

(c) No. Let F be an unpredictable function family, and define

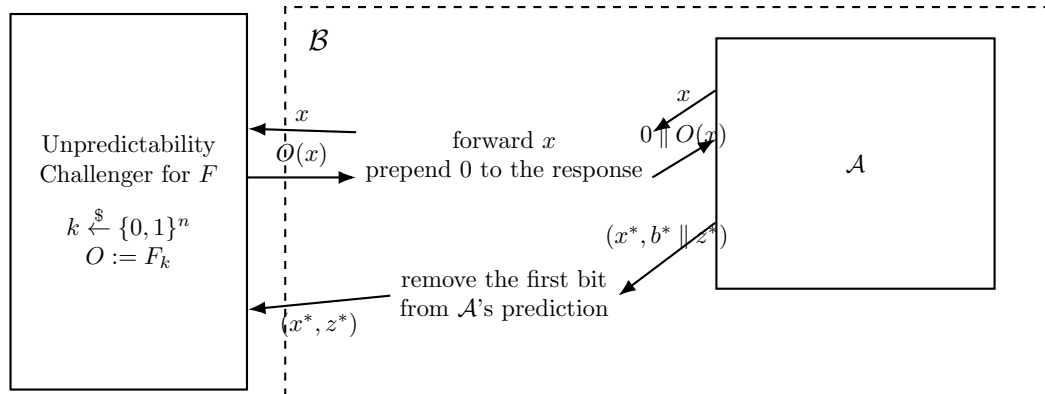
$$F'_k(x) := 0 \parallel F_k(x).$$

We claim that F' is unpredictable but not pseudorandom.

1. F' is not pseudorandom. Its first output bit is always 0. An adversary can query any input and guess that it is interacting with F' if and only if the first bit of the response is 0. When the oracle is F' , this guess is correct with probability 1; when the oracle is a random function, it is correct with probability $1/2$. Therefore, its overall success probability is

$$\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}.$$

2. F' is unpredictable. Suppose not. Then there is a PPT adversary $\mathcal{A}$ that predicts F' with non-negligible probability. We construct an adversary $\mathcal{B}$ against the unpredictability of F . Adversary $\mathcal{B}$ forwards each query x from $\mathcal{A}$ to its own oracle $O = F_k$ and returns $0 \parallel O(x)$ to $\mathcal{A}$. When $\mathcal{A}$ outputs $(x^*, b^* \parallel z^*)$, adversary $\mathcal{B}$ outputs (x^*, z^*) .



Whenever $\mathcal{A}$ wins, x^* was not previously queried and $b^* \parallel z^* = 0 \parallel F_k(x^*)$. Hence $z^* = F_k(x^*)$, so $\mathcal{B}$ also wins. Therefore,

$$\Pr[\text{Pred}_{\mathcal{B},F}(n) = 1] \geq \Pr[\text{Pred}_{\mathcal{A},F'}(n) = 1],$$

which is non-negligible. This contradicts the unpredictability of F .

■